

Question 1:

(i) Relevant Text Extraction

Methodologies:

First, we read the file/dataset provided to us in the question into a string. The data preprocessing is done on this string. The data preprocessing is done using if/else and for loop statements. Two lists “start” and “end” have been initialized. The “start” list contains the starting points of text present in between <Title> and </Title> and <Text> and </Text> respectively. The “end” list contains the ending points of text present in between <Title> and </Title> and <Text> and </Text> respectively. The moment the word <Title> is found in the string, the next word’s index is marked as the starting point of the text and the moment the word </Title> is found, the previous index is marked as the ending point of the text. Same is done for <Text> and </Text>. This is done for all the 1400 files.

Results: (Output of 5 samples)

```
+++++
Original File Content:
<DOC>
<DOCNO>
1
</DOCNO>
<TITLE>
experimental investigation of the aerodynamics of a
wing in a slipstream .
</TITLE>
<AUTHOR>
brenckman,m.
</AUTHOR>
<BIBLIO>
j. ae. scs. 25, 1958, 324.
</BIBLIO>
<TEXT>
  an experimental study of a wing in a propeller slipstream was
made in order to determine the spanwise distribution of the lift
increase due to slipstream at different angles of attack of the wing
and at different free stream to slipstream velocity ratios . the
results were intended in part as an evaluation basis for different
theoretical treatments of this problem .
  the comparative span loading curves, together with supporting
evidence, showed that a substantial part of the lift increment
produced by the slipstream was due to a /destalling/ or boundary-layer-control
effect . the integrated remaining lift increment,
after subtracting this destalling lift, was found to agree
well with a potential flow theory .
  an empirical evaluation of the destalling effects was made for
the specific configuration of the experiment .
</TEXT>
</DOC>

*****
Raw Extracted Data:
experimental investigation of the aerodynamics of a wing in a slipstream .      an experimental study of a wing in a propeller slipstream was made in
order to determine the spanwise distribution of the lift increase due to slipstream at different angles of attack of the wing and at different free st
ream to slipstream velocity ratios . the results were intended in part as an evaluation basis for different theoretical treatments of this problem .
the comparative span loading curves, together with supporting evidence, showed that a substantial part of the lift increment produced by the slipstrea
m was due to a /destalling/ or boundary layer control effect . the integrated remaining lift increment, after subtracting this destalling lift, was f
ound to agree well with a potential flow theory . an empirical evaluation of the destalling effects was made for the specific configuration of the e
xperiment .
```

```

+++++
Original File Content:
<DOC>
<DOCNO>
2
</DOCNO>
<TITLE>
simple shear flow past a flat plate in an incompressible fluid of small
viscosity .
</TITLE>
<AUTHOR>
ting-yili
</AUTHOR>
<BIBLIO>
department of aeronautical engineering, rensse laer polytechnic
institute
troy, n.y.
</BIBLIO>
<TEXT>
in the study of high-speed viscous flow past a two-dimensional body it
is usually necessary to consider a curved shock wave emitting from the
nose or leading edge of the body . consequently, there exists an inviscid
rotational flow region between the shock wave and the boundary layer
. such a situation arises, for instance, in the study of the hypersonic viscous
flow past a flat plate . the situation is somewhat different
from prandtl's classical boundary-layer problem . in prandtl's
original problem the inviscid free stream outside the boundary layer is
irrotational while in a hypersonic boundary-layer problem the inviscid
free stream must be considered as rotational . the possible effects of
vorticity have been recently discussed by ferri and libby . in the present
paper, the simple shear flow past a flat plate in a fluid of small
viscosity is investigated . it can be shown that this problem can again
be treated by the boundary-layer approximation, the only novel feature
being that the free stream has a constant vorticity . the discussion
here is restricted to two-dimensional incompressible steady flow .
</TEXT>
</DOC>

*****
Raw Extracted Data:
simple shear flow past a flat plate in an incompressible fluid of small viscosity . in the study of high speed viscous flow past a two dimensional
body it is usually necessary to consider a curved shock wave emitting from the nose or leading edge of the body . consequently, there exists an inviscid
rotational flow region between the shock wave and the boundary layer . such a situation arises, for instance, in the study of the hypersonic viscous
flow past a flat plate . the situation is somewhat different from prandtl's classical boundary layer problem . in prandtl's original problem the
inviscid free stream outside the boundary layer is irrotational while in a hypersonic boundary layer problem the inviscid free stream must be considered
as rotational . the possible effects of vorticity have been recently discussed by ferri and libby . in the present paper, the simple shear flow past
a flat plate in a fluid of small viscosity is investigated . it can be shown that this problem can again be treated by the boundary layer approximation,
the only novel feature being that the free stream has a constant vorticity . the discussion here is restricted to two dimensional incompressible
steady flow .

```

```

+++++
Original File Content:
<DOC>
<DOCNO>
3
</DOCNO>
<TITLE>
the boundary layer in simple shear flow past a flat plate .
</TITLE>
<AUTHOR>
m. b. glauert
</AUTHOR>
<BIBLIO>
department of mathematics, university of manchester, manchester,
england
</BIBLIO>
<TEXT>
the boundary-layer equations are presented for steady
incompressible flow with no pressure gradient .
</TEXT>
</DOC>

*****
Raw Extracted Data:
the boundary layer in simple shear flow past a flat plate . the boundary layer equations are presented for steady incompressible flow with no pressure
gradient .
+++++

```

```

+++++
Original File Content:
<DOC>
<DOCNO>
4
</DOCNO>
<TITLE>
approximate solutions of the incompressible laminar
boundary layer equations for a plate in shear flow .
</TITLE>
<AUTHOR>
yen,k.t.
</AUTHOR>
<BIBLIO>
j. ae. scs. 22, 1955, 728.
</BIBLIO>
<TEXT>
the two-dimensional steady boundary-layer
problem for a flat plate in a
shear flow of incompressible fluid is considered .
solutions for the boundarylayer
thickness, skin friction, and the velocity
distribution in the boundary
layer are obtained by the karman-pohlhausen
technique . comparison with
the boundary layer of a uniform flow has also
been made to show the effect of
vorticity .
</TEXT>
</DOC>

*****
Raw Extracted Data:
approximate solutions of the incompressible laminar boundary layer equations for a plate in shear flow . the two dimensional steady boundary laye
r problem for a flat plate in a shear flow of incompressible fluid is considered . solutions for the boundarylayer thickness, skin friction, and the v
elocity distribution in the boundary layer are obtained by the karman pohlhausen technique . comparison with the boundary layer of a uniform flow has
also been made to show the effect of vorticity .
+++++

```

```

Original File Content:
<DOC>
<DOCNO>
5
</DOCNO>
<TITLE>
one-dimensional transient heat conduction into a double-layer
slab subjected to a linear heat input for a small time
internal .
</TITLE>
<AUTHOR>
wasserman,b.
</AUTHOR>
<BIBLIO>
j. ae. scs. 24, 1957, 924.
</BIBLIO>
<TEXT>
analytic solutions are presented for the transient heat conduction
in composite slabs exposed at one surface to a
triangular heat rate . this type of heating rate may occur, for
example, during aerodynamic heating .
</TEXT>
</DOC>

*****
Raw Extracted Data:
one dimensional transient heat conduction into a double layer slab subjected to a linear heat input for a small time internal . analytic solution
s are presented for the transient heat conduction in composite slabs exposed at one surface to a triangular heat rate . this type of heating rate may
occur, for example, during aerodynamic heating .
+++++

```

(ii) Preprocessing

Methodologies:

The text was converted to lowercase by using `.lower()`. Tokenization was performed using `word_tokenize()`. Alternatively we could have also performed tokenization by using space as a separator. We obtained a list of punctuations using `string.punctuation()` and then we replaced those punctuations with spaces

in the final string. For removing stopwords, we obtained a list of stopwords using stopwords.word(“english”). Then using a for loop we appended all the words except the stopwords in another list. There were no blank space tokens as they were removed through our method of tokenization.

Results:(Output of 5 samples)

```
+++++
Before Preprocessing:
 experimental investigation of the aerodynamics of a wing in a slipstream . an experimental study of a wing in a propeller slipstream was made in
 order to determine the spanwise distribution of the lift increase due to slipstream at different angles of attack of the wing and at different free st
ream to slipstream velocity ratios . the results were intended in part as an evaluation basis for different theoretical treatments of this problem .
 the comparative span loading curves, together with supporting evidence, showed that a substantial part of the lift increment produced by the slipstrea
m was due to a /destalling/ or boundary layer control effect . the integrated remaining lift increment, after subtracting this destalling lift, was f
ound to agree well with a potential flow theory . an empirical evaluation of the destalling effects was made for the specific configuration of the e
xperiment .
*****
After Preprocessing: (Comma Separated Tokens)
experimental, investigation, aerodynamics, wing, slipstream, experimental, study, wing, propeller, slipstream, made, order, determine, spanwise, distr
ibution, lift, increase, due, slipstream, different, angles, attack, wing, different, free, stream, slipstream, velocity, ratios, results, intended, p
art, evaluation, basis, different, theoretical, treatments, problem, comparative, span, loading, curves, together, supporting, evidence, showed, subst
antial, part, lift, increment, produced, slipstream, due, destalling, boundary, layer, control, effect, integrated, remaining, lift, increment, subtra
cting, destalling, lift, found, agree, well, potential, flow, theory, empirical, evaluation, destalling, effects, made, specific, configuration, exper
iment, +++++
```

```
Before Preprocessing:
 simple shear flow past a flat plate in an incompressible fluid of small viscosity . in the study of high speed viscous flow past a two dimensional
 body it is usually necessary to consider a curved shock wave emitting from the nose or leading edge of the body . consequently, there exists an invis
cid rotational flow region between the shock wave and the boundary layer . such a situation arises, for instance, in the study of the hypersonic visc
ous flow past a flat plate . the situation is somewhat different from prandtl's classical boundary layer problem . in prandtl's original problem the
inviscid free stream outside the boundary layer is irrotational while in a hypersonic boundary layer problem the inviscid free stream must be consider
ed as rotational . the possible effects of vorticity have been recently discussed by ferri and libby . in the present paper, the simple shear flow p
ast a flat plate in a fluid of small viscosity is investigated . it can be shown that this problem can again be treated by the boundary layer approxi
mation, the only novel feature being that the free stream has a constant vorticity . the discussion here is restricted to two dimensional incompressi
ble steady flow .
*****
After Preprocessing: (Comma Separated Tokens)
simple, shear, flow, past, flat, plate, incompressible, fluid, small, viscosity, study, high, speed, viscous, flow, past, two, dimensional, body, usua
lly, necessary, consider, curved, shock, wave, emitting, nose, leading, edge, body, consequently, exists, inviscid, rotational, flow, region, shock, w
ave, boundary, layer, situation, arises, instance, study, hypersonic, viscous, flow, past, flat, plate, situation, somewhat, different, prandtl, class
ical, boundary, layer, problem, prandtl, original, problem, inviscid, free, stream, outside, boundary, layer, irrotational, hypersonic, boundary, laye
r, problem, inviscid, free, stream, must, considered, rotational, possible, effects, vorticity, recently, discussed, ferri, libby, present, paper, sim
ple, shear, flow, past, flat, plate, fluid, small, viscosity, investigated, shown, problem, treated, boundary, layer, approximation, novel, feature, f
ree, stream, constant, vorticity, discussion, restricted, two, dimensional, incompressible, steady, flow, +++++
```

```
Before Preprocessing:
 the boundary layer in simple shear flow past a flat plate . the boundary layer equations are presented for steady incompressible flow with no press
ure gradient .
*****
After Preprocessing: (Comma Separated Tokens)
boundary, layer, simple, shear, flow, past, flat, plate, boundary, layer, equations, presented, steady, incompressible, flow, pressure, gradient, +++
++++
```

```
Before Preprocessing:
 approximate solutions of the incompressible laminar boundary layer equations for a plate in shear flow . the two dimensional steady boundary laye
r problem for a flat plate in a shear flow of incompressible fluid is considered . solutions for the boundarylayer thickness, skin friction, and the v
elocity distribution in the boundary layer are obtained by the karman pohlhausen technique . comparison with the boundary layer of a uniform flow has
also been made to show the effect of vorticity .
*****
After Preprocessing: (Comma Separated Tokens)
approximate, solutions, incompressible, laminar, boundary, layer, equations, plate, shear, flow, two, dimensional, steady, boundary, layer, problem, f
lat, plate, shear, flow, incompressible, fluid, considered, solutions, boundarylayer, thickness, skin, friction, velocity, distribution, boundary, lay
er, obtained, karman, pohlhausen, technique, comparison, boundary, layer, uniform, flow, also, made, show, effect, vorticity, +++++
```

```
Before Preprocessing:
 one dimensional transient heat conduction into a double layer slab subjected to a linear heat input for a small time internal . analytic solution
s are presented for the transient heat conduction in composite slabs exposed at one surface to a triangular heat rate . this type of heating rate may
occur, for example, during aerodynamic heating .
*****
After Preprocessing: (Comma Separated Tokens)
one, dimensional, transient, heat, conduction, double, layer, slab, subjected, linear, heat, input, small, time, internal, analytic, solutions, presen
ted, transient, heat, conduction, composite, slabs, exposed, one, surface, triangular, heat, rate, type, heating, rate, may, occur, example, aerodynam
ic, heating, +++++
```

Question 2:

Methodologies:

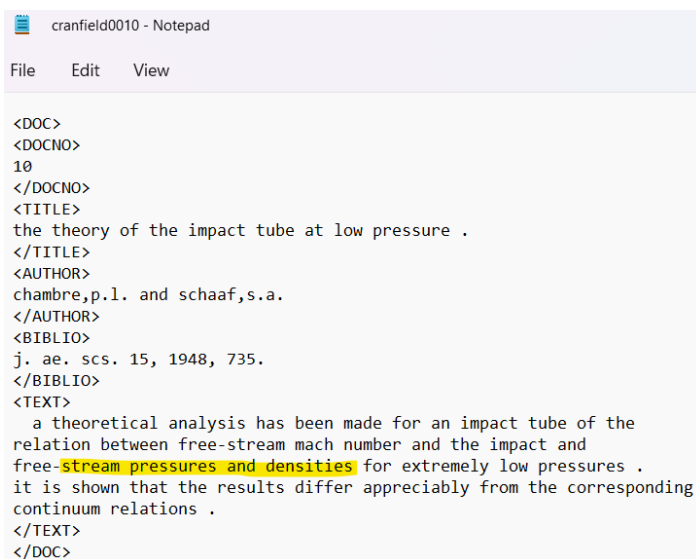
For Unigram Inverted Index, first we perform the preprocessing of all the data in the files and remove the duplicate occurrences of words in each document as they hold no value when creating a unigram inverted index. Then a dictionary with a token as the key and a list as the value is initialized. Using a for loop we go over each token of every document, the token is the key and the value is a sorted list of all the document numbers in which the key(token) is present. If the token is already in the dictionary then we append the document number to the list associated with it, if the token is not present in the dictionary then we add it in the dictionary along with a new list containing the respective document number.

Merging algorithm was used to calculate the intersection and union of the boolean queries. We also maintained a universal set to perform the “not” boolean query. The comparisons calculated in the code are the comparisons made during merging.

Pickle module was used to load and save a unigram inverted index.

Results:

```
+++++
Original Input Query: stream and pressures and densities
Simplified Input Query: stream and pressures and densities
Number of Documents: 1
Document Names:
    1. Data/CSE508_Winter2023_Dataset/cranfield0010
Number of comparisons for fetching result: 311
+++++
```



```
cranfield0010 - Notepad
File Edit View

<DOC>
<DOCNO>
10
</DOCNO>
<TITLE>
the theory of the impact tube at low pressure .
</TITLE>
<AUTHOR>
chambre,p.l. and schaaf,s.a.
</AUTHOR>
<BIBLIO>
j. ae. scs. 15, 1948, 735.
</BIBLIO>
<TEXT>
a theoretical analysis has been made for an impact tube of the
relation between free-stream mach number and the impact and
free-stream pressures and densities for extremely low pressures .
it is shown that the results differ appreciably from the corresponding
continuum relations .
</TEXT>
</DOC>
```

Question 3:

(i) Bigram inverted index

Methodologies:

For the Bigram Inverted Index, our approach was very similar to the unigram index. However, this time, instead of using individual words as tokens, we used the pair of a word and both its neighbors as tokens. This approach of tokenizing is implemented to the Inverted Index, and to the input query as well.

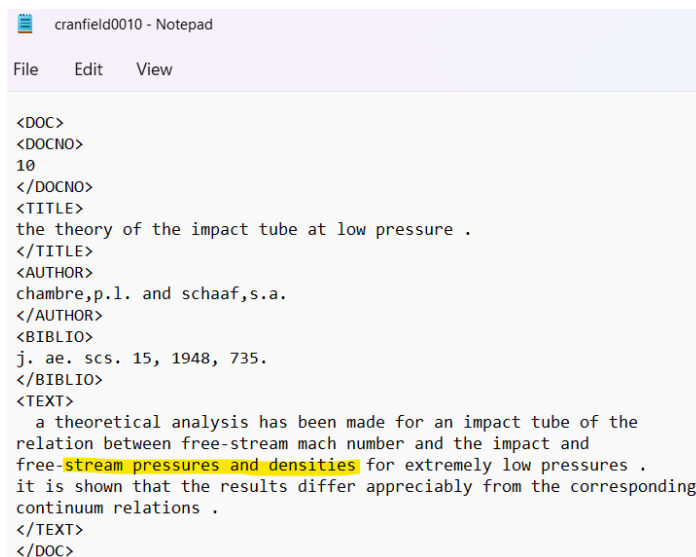
For searching phrases longer than two words, we break these phrases into tokens containing two words, and then concatenate these tokens using the 'and' operator to generate a query which can then be searched.

Merging algorithm was used to calculate the intersection of the boolean queries. As explained in the previous paragraph, we only concatenated the tokens with and operator, hence, we only had to implement intersection using merge.

Pickle module was used to load and save a unigram inverted index.

Results:

```
+++++
Search Phrase: stream pressures densities
Input Query: stream pressures and pressures densities
Number of Documents Retrieved: 1
The following documents contain your search phrase:
    1. Data/CSE508_Winter2023_Dataset/cranfield0010
+++++
```



```
cranfield0010 - Notepad
File Edit View

<DOC>
<DOCNO>
10
</DOCNO>
<TITLE>
the theory of the impact tube at low pressure .
</TITLE>
<AUTHOR>
chambre,p.l. and schaaf,s.a.
</AUTHOR>
<BIBLIO>
j. ae. scs. 15, 1948, 735.
</BIBLIO>
<TEXT>
a theoretical analysis has been made for an impact tube of the
relation between free-stream mach number and the impact and
free-stream pressures and densities for extremely low pressures .
it is shown that the results differ appreciably from the corresponding
continuum relations .
</TEXT>
</DOC>
```

ii) Positional index:

Methodologies:

For implementing positional indexes, we chose to go with a dictionary with key as a 'token' and value as another child dictionary. This child dictionary contains document numbers as keys and a sorted list of positions of the parent dictionaries key(the tokens) as the value.

We chose to go for this structure of implementation as it allows use to perform much faster searches.

Whenever a new document is added to this index, it will pre-process the document and create tokens. For each token, it will check if the token exists in the dictionary, if it does, then it will check if the current document number exists in the dictionary of that particular token. If that is also true, then it means that a list containing positions of this token within the current document exists, and the algorithm will append the position of the token in this list.

If the token exists but the document doesn't, then the document will be added to the dictionary of the token as a key and the corresponding value will be a new list of positions.

For searching we implemented an optimized, binary search based searching approach which helps us perform quick searches. When searching for a phrase, we first search for all occurrences of the first word. Each occurrence consists of a document number and the position in that document. We then search for the occurrence of the second word, in the same document, at the next position. If it exists, we then repeat the same for the third word. Otherwise, we choose a different occurrence of the first word.

Whenever all words match, i.e. exact match, we store the document of the first word in our output list.

Pickle module was used to load and save a unigram inverted index.

We have made many optimizations to boost the performance of this positional index.

Results:

```
+++++
Search Phrase: stream pressures densities
Number of Documents Retrieved: 1
The following documents contain your search phrase:
    1. Data/CSE508_Winter2023_Dataset/cranfield0010
+++++
```



```
cranfield0010 - Notepad
File Edit View

<DOC>
<DOCNO>
10
</DOCNO>
<TITLE>
the theory of the impact tube at low pressure .
</TITLE>
<AUTHOR>
chambre,p.l. and schaaf,s.a.
</AUTHOR>
<BIBLIO>
j. ae. scs. 15, 1948, 735.
</BIBLIO>
<TEXT>
a theoretical analysis has been made for an impact tube of the
relation between free-stream mach number and the impact and
free-stream pressures and densities for extremely low pressures .
it is shown that the results differ appreciably from the corresponding
continuum relations .
</TEXT>
</DOC>
```

iii) Comparison between Positional Inverted Index and Bigram Inverted Index

Using our results, we made the following observations. We have also provided the results obtained from code to support our observations:

1. The creation time of the Bigram Inverted Index is almost similar to the creation time of the Positional Inverted Index. Although Bigram might be faster for larger datasets due to its less complex structure.

Creation Time Comparison:

Bigram Inverted Index Creation Time: 1.238358497619629

Positional Inverted Index Creation Time: 1.2803966999053955

-
2. Bigram Inverted Index cannot be used to search single words in a document.
 3. Positional Inverted Index can be used to search single words in a document.

Single Word Query Comparison:

Search Phrase: experimental

Number of Documents Retrieved Using Bigram Index: 0

No matches found!

4. Bigram Inverted Index performs phrase query searching significantly faster. Although it may give false positive outputs in some cases.

5. Even though Positional Inverted Index performs phrase query searching significantly slower than bigram, It will never give false positives outputs in any case.

Phrase Query Comparison:

Search Phrase: boundary layer equations

Input Query: boundary layer and layer equations

Number of Documents Retrieved Using Bigram Index: 41

Bigram Inverted Index Query Time: 0.0005202293395996094

Positional Inverted Index Query Time: 0.001561880111694336